SOC Lab Report: Lateral Movement Detection using Splunk

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1 Abstract

This report demonstrates detection of lateral movement in a SOC environment using Splunk. After gaining access, attackers move across systems using remote services such as RDP. This scenario simulates remote login attempts and detects them using Windows event logs.

2 Objective

To detect lateral movement by identifying remote login activity using Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Lateral Movement
- Technique: Remote Services (T1021)

5 Attack Simulation

Lateral movement was simulated using RDP login attempts from Kali Linux.

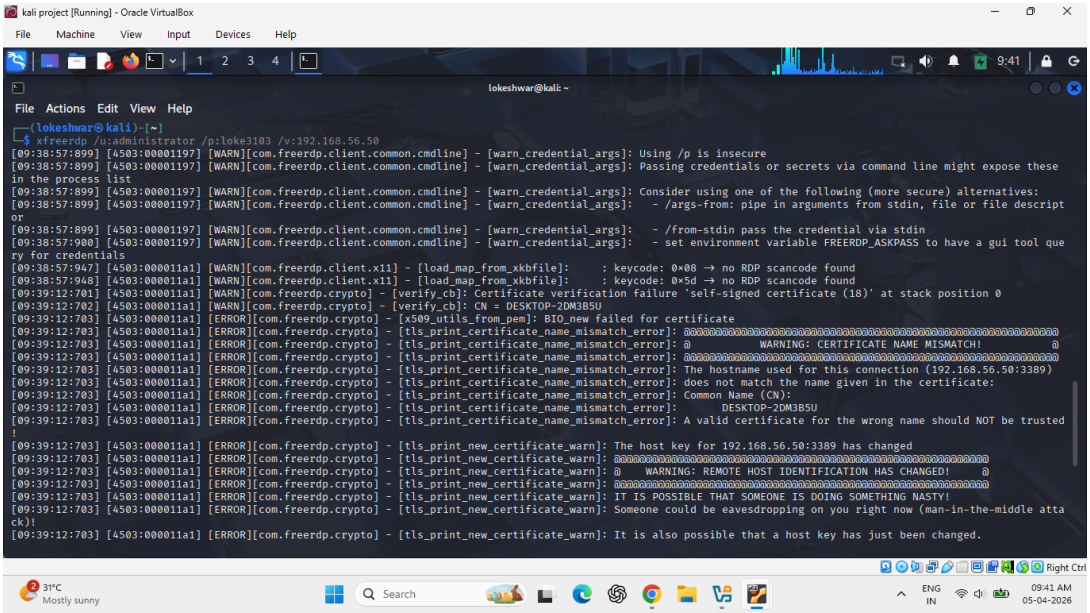
5.1 Command Used

```
xfreerdp /u:administrator /p:Password123 /v:192.168.56.50
```

5.2 Explanation

The command attempts a remote desktop login using compromised credentials.

5.3 Screenshot


kali project [Running] - Oracle VirtualBox
File Machine View Input Devices Help
lokeshwar@kali: ~
File Actions Edit View Help
┌──(lokeshwar㉿kali)-[~]
└─$ xfreerdp /u:administrator /p:loke3103 /v:192.168.56.50
[09:38:57:899] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Using /p is insecure
[09:38:57:899] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Passing credentials or secrets via command line might expose these in the process list
[09:38:57:899] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Consider using one of the following (more secure) alternatives:
[09:38:57:899] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /args-from: pipe in arguments from stdin, file or file descript or
[09:38:57:899] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /from-stdin pass the credential via stdin
[09:38:57:900] [4503:00001197] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - set environment variable FREERDP_ASKPASS to have a gui tool que ry for credentials
[09:38:57:947] [4503:000011a1] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP scancode found
[09:38:57:948] [4503:000011a1] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5d → no RDP scancode found
[09:39:12:701] [4503:000011a1] [WARN][com.freerdp.crypto] - [verify_cb]: Certificate verification failure 'self-signed certificate (18)' at stack position 0
[09:39:12:702] [4503:000011a1] [WARN][com.freerdp.crypto] - [verify_cb]: CN = DESKTOP-2DM3B5U
[09:39:12:703] [4503:000011a1] [ERROR][com.freerdp.crypto] - [x509_utils_from_pem]: BIO_new failed for certificate
[09:39:12:703] [4503:000011a1] [ERROR][com.freerdp.crypto] - [tls_print_certificate_name_mismatch_error]: @@@@

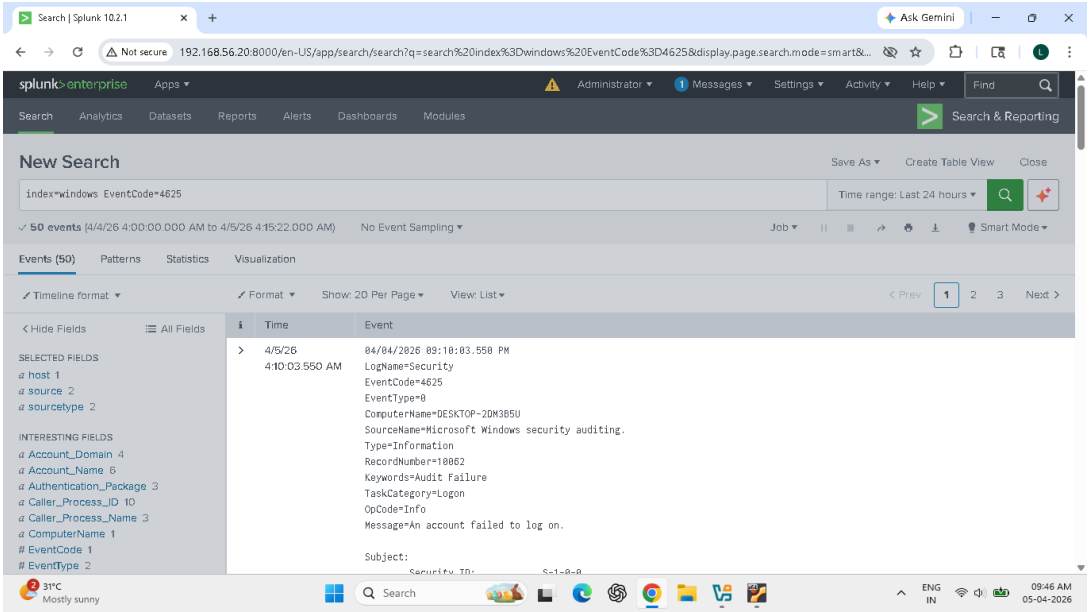


Figure 2: Detection of Remote Login Activity

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Source_Network_Address
```

7.2 Findings

- Remote login detected from suspicious IP
- Same user accessed system remotely
- Possible lateral movement attempt

7.3 Screenshot

8 Response Actions

Immediate steps were taken to prevent further lateral movement.

- Disabled compromised account
- Blocked attacker IP
- Disabled RDP access

8.1 Commands Used

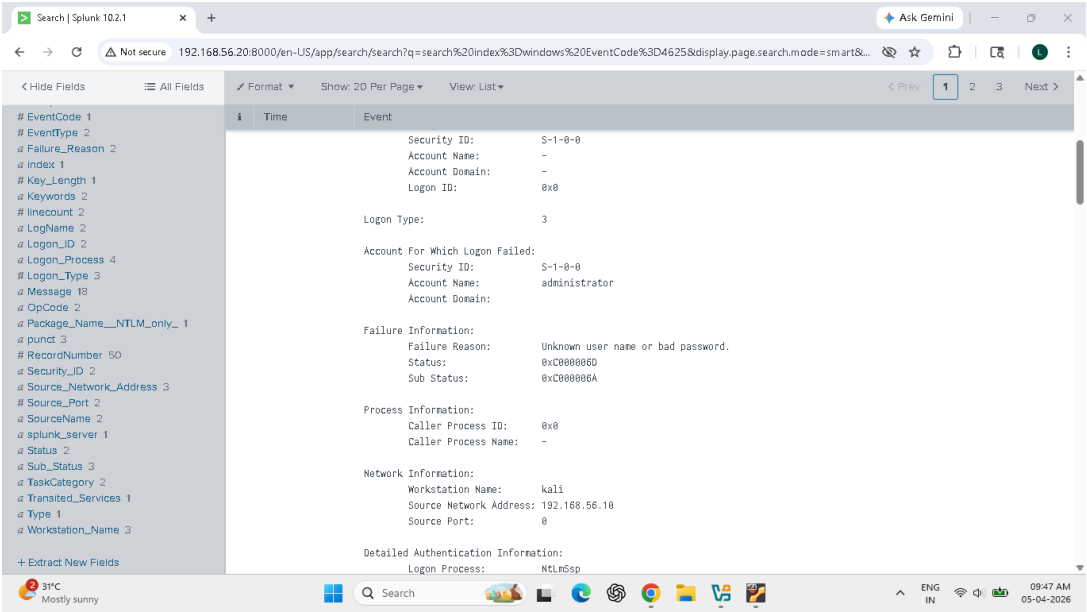


Figure 3: Detection of Remote Login Activity

```
net user <username> /active:no
New-NetFirewallRule -DisplayName "Block-Attacker" -Direction Inbound -Remo
Set-ItemProperty -Path "HKLM:\System\CurrentControlSet\Control\Terminal-Ser
```

8.2 Explanation

The compromised account was disabled, attacker IP was blocked, and RDP access was turned off to prevent further intrusion.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

A remote login was detected from a suspicious IP, indicating lateral movement using compromised credentials.

9.2 Impact

The attacker could access multiple systems within the network.

9.3 Conclusion

The lateral movement was detected and mitigated using Splunk and appropriate response actions.

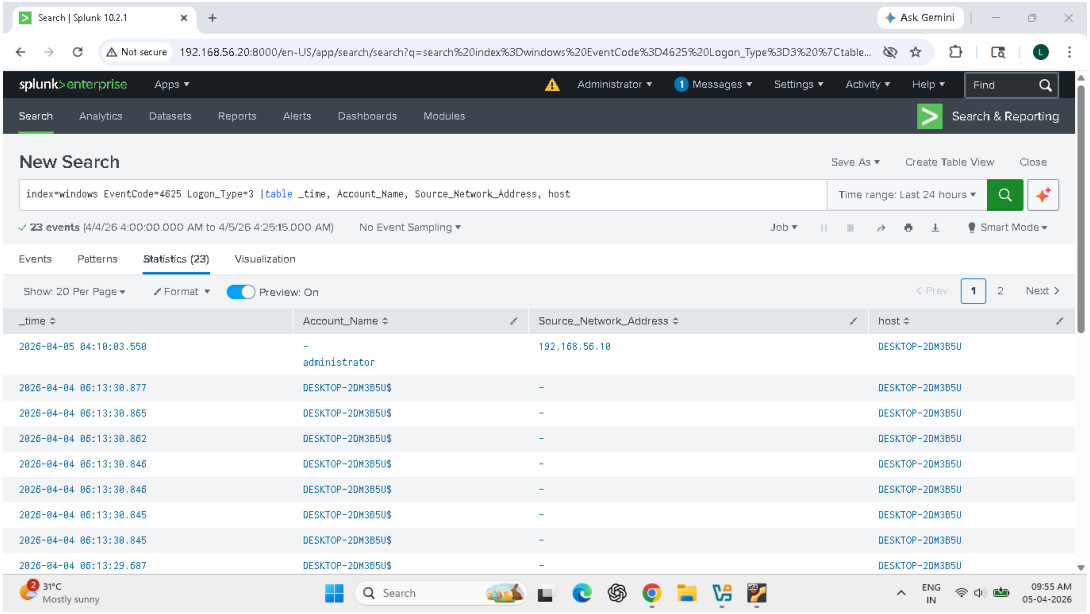


Figure 4: Timeline showing lateral movement activity

10 Conclusion

This scenario highlights the importance of monitoring remote login activities to detect lateral movement within a network.

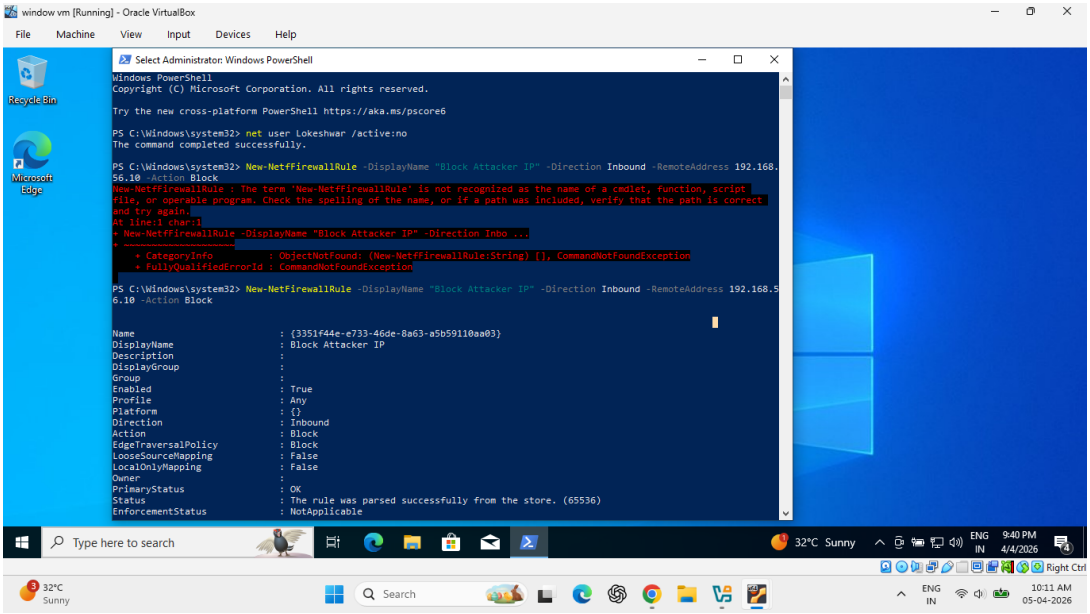


Figure 5: Response Actions - Blocking Lateral Movement